

StockSense – Technical Learning Notes

Module: Hardware Core Concepts & Signal Processing

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1. Load Cells, Strain Gauges & Wheatstone Bridge Dynamics

Strain Gauge Operating Principle

A strain gauge consists of an insulated flexible backing supporting a metallic foil pattern. When subject to mechanical force or weight, physical deformation alters its internal resistance (ΔR):

- **Tension (Stretching):** Gauge wire lengthens and thins → Electrical resistance (R) increases.
- **Compression (Squeezing):** Gauge wire shortens and thickens → Electrical resistance (R) decreases.

Wheatstone Bridge Configuration

Because fractional changes in resistance ($\Delta R/R$) are extremely minute, four strain gauges are configured in a Wheatstone bridge circuit to convert differential resistance shifts into a readable millivolt output (V_{out}):

$$V_{out} = V_E \cdot \left(\frac{R_1}{R_1 + R_2} - \frac{R_4}{R_3 + R_4} \right) \quad (1)$$

Where:

- V_E is the excitation voltage applied to the bridge (typically 5V or 3.3V).
- R_1, R_2, R_3, R_4 represent the four active strain gauge resistors on the load cell body.
- When no load is applied, the bridge is balanced ($V_{out} = 0$ mV). Load application unbalances the bridge, outputting a differential voltage proportional to weight.

2. The HX711 24-Bit ADC Module

Microcontrollers such as the ESP32 feature onboard 12-bit ADCs that lack the resolution and low-noise front-end needed to measure millivolt-level differential signals accurately. The HX711 chip provides two key hardware functions:

1. **Programmable Gain Amplifier (PGA):** Amplifies tiny millivolt inputs from Channel A by a selectable hardware gain of **128** or **64**.
2. **24-Bit Sigma-Delta ADC:** Digitizes the amplified analog signal into a 24-bit unsigned/signed integer representation:

$$\text{Resolution Range} = 0 \text{ to } (2^{24} - 1) = 0 \text{ to } 16,777,215 \text{ counts}$$

Hardware Interface & Control Lines

- **Bridge Terminals:** $E+$, $E-$ (Excitation Voltage Output), $A+$, $A-$ (Differential Analog Input).
- **Microcontroller Interface:**
 - PD_SCK (Power Down & Serial Clock): Clock pulses generated by the ESP32 GPIO.
 - DOUT (Data Out): Pulled LOW when data is ready; bits are shifted out on PD_SCK clock edges.

3. Mathematical Formulas for Weight & Quantity Derivation

Firmware Weight Calibration Equation

To convert raw 24-bit ADC counts into a physical calibrated mass (grams or kilograms), the edge firmware executes:

$$\text{Measured Weight} = \frac{\text{Raw ADC Read Value} - \text{Tare Offset}}{\text{Scale Calibration Factor}} \quad (2)$$

Where:

- **Tare Offset:** The zero-load ADC baseline reading (compensates for empty bin platform weight).
- **Scale Calibration Factor:** Digital counts per unit weight, derived during device calibration using a known test mass.

Discrete Item Quantity Calculation

Once total net weight is determined, item count is calculated in firmware or backend analytics as:

$$\text{Item Quantity} = \text{round} \left(\frac{\text{Measured Weight}}{\text{Unit Weight of Single Item}} \right) \quad (3)$$

Where:

- Unit Weight of Single Item is stored in the database for the active RFID/QR tag linked to that bin.

4. Next Study Topics (Upcoming Week 2 Prep)

- **RFID Identification Protocols:** SPI and I²C interface timing for MFRC522 and PN532 RFID readers.
- **MQTT Architecture & Transport:** Design of topic trees (`stocksense/site_id/bin_id/telemetry`), QoS 0 vs QoS 1 guarantees, and JSON payload definitions.

5. RFID Reader Protocols (MFRC522 & PN532)

RFID uses electromagnetic fields to identify tags automatically. In StockSense, a passive tag on each bin links its Unique Identifier (UID) to a product record in the database.

```
[Passive RFID Tag on Bin] ~ 13.56 MHz HF field ~ [Reader] --SPI/I2C--> ESP32
```

Key Hardware Comparison

Feature	MFRC522	PN532
Frequency	13.56 MHz (HF)	13.56 MHz (HF)
Supported tags	MIFARE Classic / Ultralight	MIFARE, NTAG, NFC Forum tags
Interfaces	SPI (default), I ² C, UART	SPI, I ² C (DIP switch), HSU (UART)
NFC support	Card reading only	Full NFC (read/write, peer-to-peer, card emulation)
Best for	Simple tag UID reading per bin	Tag encryption or future phone scanning

Communication Protocols

SPI (Serial Peripheral Interface)

- 4 bus wires: MOSI, MISO, SCK and SS/SDA (chip select).
- Very fast (10 MHz+ clock).
- Devices share MOSI, MISO and SCK, but each needs its own SS pin on the ESP32.

I²C (Inter-Integrated Circuit)

- 2 wires only: SDA (data) and SCL (clock).
- 100 kHz (standard) or 400 kHz (fast mode).
- Several devices (e.g. RFID + temperature/humidity sensor) share the same 2 pins using unique addresses (e.g. 0x28, 0x53).

6. MQTT Protocol Architecture

MQTT (Message Queuing Telemetry Transport) is a lightweight publish/subscribe protocol designed for low-bandwidth, high-latency or unreliable networks.

```
[ESP32 Node] --publish JSON--> [Mosquitto / EMQX Broker] --subscribe--> [Dashboard / Cloud DB]
```

Publish/Subscribe Model

- **Publisher (ESP32):** sends data under named channels called topics.
- **Broker (Mosquitto / EMQX):** receives messages, filters by topic and routes them to subscribers.
- **Subscriber (backend / dashboard):** listens to topics to update stock data in real time.

Quality of Service (QoS) Levels

Level	Guarantee	Use in StockSense
QoS 0	At most once ("fire and forget"); may be lost if Wi-Fi drops.	Routine continuous weight updates
QoS 1	At least once; delivery confirmed by PUBACK.	Stock events: item removal, theft and low-stock alerts
QoS 2	Exactly once via 4-step handshake; highest bandwidth and power cost.	Rarely needed

MQTT Topic Hierarchy & JSON Payload

stocksense/<site_id>/bin/<bin_id>/<data_type>

- Telemetry: stocksense/site_01/bin/bin_01/telemetry
- Stock alert: stocksense/site_01/bin/bin_01/alerts
- Status / LWT: stocksense/site_01/bin/bin_01/status

Telemetry payload example:

```
{
  "device_id": "ESP32_NODE_01",
  "bin_id": "BIN_A1",
  "tag_uid": "A4B23C99",
  "weight_g": 1450.5,
  "unit_weight_g": 10.0,
  "quantity": 145,
  "temperature_c": 26.4,
  "humidity_pct": 65.2,
  "timestamp": 1728060960
}
```